Linearized Dual Smoothing Augmented Lagrangian methods for Nonconvex Constrained Stochastic Optimization

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1 Introduction

We consider the following nonconvex stochastic optimization problem with nonlinear equality constraints:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathcal{X}} \quad & F(\boldsymbol{x}) := f(\boldsymbol{x}) + g(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{c}(\boldsymbol{x}) = \mathbf{0}. \end{aligned} \tag{\textbf{P}}$$

where $f(\boldsymbol{x}) = \mathbb{E}_\xi[\mathbf{f}(\boldsymbol{x},\xi)]$ is a smooth nonconvex function with ξ a random variable on a probability space Ξ independent of $\boldsymbol{x}$, and $g:\mathbb{R}^n \to \mathbb{R}\cup\{+\infty\}$ is a proper closed proximal-friendly function. The mapping $\boldsymbol{c}:\mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable. The set $\mathcal{X}\subseteq\mathbb{R}^n$ is assumed to be nonempty and closed. The feasible set is assumed to be nonempty and F is lower bounded by $F^* > -\infty$. Exact evaluation of ∇f is often expensive or impractical, and thus we assume only access to stochastic gradients of f at queried points.

1.1 Related Work

Deterministic nonconvex constrained optimization. For deterministic nonconvex problems with nonlinear equality constraints, first-order methods with non-asymptotic complexity guarantees have been studied extensively. Proximal point methods [4, 20] achieve $\mathcal{O}(\varepsilon^{-3})$ iteration complexity for smooth objectives via strongly convex subproblems, and Boob et al. [5] further attain $\mathcal{O}(\varepsilon^{-2})$ by constructing quadratic models with adaptive constraint levels. Among penalty and augmented Lagrangian methods (ALMs), most approaches require the penalty parameter to grow as $\Omega(\varepsilon^{-1})$, yielding $\mathcal{O}(\varepsilon^{-3})$ iteration complexity [15, 13]. Xie and Wright [24] achieve $\mathcal{O}(\varepsilon^{-2})$ outer-loop complexity with a proximal ALM under a constant merit parameter, but the algorithm is double-loop and the merit parameter must satisfy $\rho = \Omega(\varepsilon^{-\eta})$ for some $\eta\in(0,1)$, making it ε-dependent in theory. SQP methods [9, 6] also attain $\mathcal{O}(\varepsilon^{-2})$ complexity.

Stochastic nonconvex constrained optimization. We organize the literature by the two canonical settings: *semi-stochastic* (stochastic objective, deterministic constraints) and *fully stochastic* (both objective and constraints involve stochastic oracles).

Semi-stochastic problems. Without additional structure on the noise, proximal point [4, 20], penalty [14], and SQP [9] methods all achieve $\mathcal{O}(\varepsilon^{-4})$ oracle complexity in stochastic gradient evaluations, matching the $\mathcal{O}(\varepsilon^{-4})$ lower bound for unconstrained stochastic optimization [3]. Under a mean-squared smoothness (MSS) condition, variance reduction techniques—STORM [10] and SPIDER [11]—improve the complexity to $\mathcal{O}(\varepsilon^{-3})$. Representative results include the variance-reduced linearized ALM of Shi et al. [23], the truncated variance reduction approach of Lu et al. [18], and the inexact ALM of Li et al. [15], each attaining $\mathcal{O}(\varepsilon^{-3})$ oracle complexity under MSS for nonlinear constraints.

Fully stochastic problems. When constraint evaluations are also stochastic, complexity results are uniformly higher. Proximal point methods [4, 20] incur $\mathcal{O}(\varepsilon^{-6})$ total oracle complexity due to the cost of solving stochastic strongly convex subproblems. Penalty-based ALMs employing STORM-type variance

reduction [1, 7] face a large merit parameter $\beta = \Theta(\varepsilon^{-1})$, which amplifies the variance of the stochastic gradient and yields $\mathcal{O}(\varepsilon^{-5})$ complexity even under MSS. Exact penalty methods combined with SPIDER [17] achieve $\mathcal{O}(\varepsilon^{-4})$ (sub)gradient and $\mathcal{O}(\varepsilon^{-6})$ constraint-evaluation complexity for nonsmooth problems. Cui et al. [8] recently propose an adaptive exact penalty method for smooth *equality*-constrained problems, obtaining $\mathcal{O}(\varepsilon^{-3})$ gradient and $\mathcal{O}(\varepsilon^{-5})$ constraint-evaluation complexity; however, the analysis is confined to equality constraints and relies on an exact penalty reformulation. Stochastic SQP for equality-constrained fully stochastic problems [21] currently achieves only $\mathcal{O}(\varepsilon^{-8})$.

Linearly constrained nonconvex problem *Deterministic objective.* The augmented Lagrangian (AL) framework, introduced by Hestenes [**?**] and Powell [**?**] and systematically developed by Bertsekas [**?**], forms the algorithmic backbone for constrained optimization. For nonconvex problems, early global convergence results were established for ADMM-type methods under boundedness assumptions on the dual iterates [**?** **?** **?**]. From a complexity standpoint, Hajinezhad and Hong [**?**] proposed a perturbed proximal primal-dual algorithm achieving $O(\varepsilon^{-4})$ iteration complexity, but requiring a feasible initialization. Kong et al. [**?** **?**] developed double-loop proximal AL methods attaining $O(\varepsilon^{-3})$ and further $O(\varepsilon^{-2.5})$ complexity under Slater's condition; however, these require solving subproblems in an inner loop, increasing implementation overhead. The first single-loop method achieving $O(\varepsilon^{-2})$ iteration complexity was proposed by Zhang and Luo [**?** **?**] via the SProx-ALM framework, exploiting Hoffman's error bound for the linear constraints. More recently, Kim [**?**] introduced the Proximal-Perturbed Lagrangian (PPAL) framework, which uses artificial variables and a dual smoothing term ("false penalty") to enable a single-loop algorithm with $O(\varepsilon^{-2})$ complexity. A distinctive feature of PPAL is that it works with a *fixed* penalty parameter and does not require feasible initialization, Slater's condition, or full column rank of A.

Stochastic objective. When $f(x) = \mathbb{E}_\xi[F(x;\xi)]$ is an expectation function accessible only via stochastic oracles, the analysis becomes substantially more challenging due to gradient noise accumulation. Shi, Wang, and Wang [**?**] proposed MLALM, a single-loop linearized AL method incorporating a recursive momentum (STORM-type) technique for stochastic gradient estimation. Under an unbounded increasing penalty schedule, MLALM achieves $O(\varepsilon^{-4})$ oracle complexity to find an ε-stationary point or ε-KKT point; this improves to $O(\varepsilon^{-3})$ only when the initial iterate is *nearly feasible* within a prescribed accuracy. Alacaoglu and Wright [**?**] systematically studied single-loop variance-reduced AL and quadratic penalty algorithms across three constraint regimes: for *linear deterministic* constraints, they achieved $\widetilde{O}(\varepsilon^{-3})$ sample complexity using a constant penalty parameter and variance reduction, matching the lower bound for unconstrained stochastic nonconvex optimization; for *nonlinear deterministic* constraints, $\widetilde{O}(\varepsilon^{-4})$; and for *stochastic nonlinear* constraints, $\widetilde{O}(\varepsilon^{-5})$. Notably, they explicitly identified as an open problem the question of whether ALM with a *fixed* penalty parameter (independent of ε) can achieve $O(\varepsilon^{-3})$ complexity for nonlinear equality constraints. Lu, Mei, and Xiao [**?**] addressed this gap partially via truncated recursive and Polyak momentum schemes, establishing $\widetilde{O}(\varepsilon^{-3})$ sample complexity together with *high-probability* (ε-surely feasible) guarantees under an error bound condition; however, their analysis requires gradient clipping and relies on the error bound parameter $\theta \geq 1$, with the increasing penalty schedule remaining necessary. In the PPAL setting with stochastic objectives, Zuo, Wang, and Wang [**?**] proposed AdaSSP, an adaptive single-loop stochastic penalty method with high-probability complexity analysis; this method also employs an unbounded penalty sequence.

ALMs with fixed penalty parameter. A fundamental challenge in ALM design for nonlinear constraints is that standard convergence analysis requires the penalty parameter to scale as $\Omega(\varepsilon^{-1})$. This growth inflates the Lipschitz constant of the augmented Lagrangian and forces the step size to shrink proportionally, degrading the stochastic oracle complexity. For polyhedral (linear) constraints, the Luo–Pang global error bound [19] decouples feasibility control from the penalty magnitude. Exploiting this structure, Zhang and Luo [25] achieve $\mathcal{O}(\varepsilon^{-2})$ iteration complexity for deterministic nonconvex problems under a genuinely constant penalty. In the stochastic setting, Alacaoglu and Wright [2] combine a fixed penalty with variance reduction to attain the optimal $\mathcal{O}(\varepsilon^{-3})$ sample complexity for linear equality constraints; for nonlinear constraints, their single-loop algorithm requires an increasing penalty and achieves

only $\mathcal{O}(\varepsilon^{-4})$. Huang et al. [12] extend the fixed-penalty result to linear inequality constraints via Moreau envelope smoothing, also reaching $\mathcal{O}(\varepsilon^{-3})$. For general nonlinear constraints, existing stochastic methods either require an increasing penalty parameter [16, 2], yielding $\mathcal{O}(\varepsilon^{-4})$ oracle complexity, or introduce dual momentum to partially compensate, achieving $\mathcal{O}(\varepsilon^{-3.5})$ [22]. To date, no single-loop ALM with a genuinely constant penalty parameter has been shown to achieve $\mathcal{O}(\varepsilon^{-3})$ oracle complexity for nonlinear nonconvex constraints in the stochastic setting.

The proposed Perturbed ALM addresses this gap. By introducing a dual regularization term $-\frac{\beta}{2}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$ into the augmented Lagrangian, the effective penalty parameter reduces to $\rho = \alpha/(1+\alpha\beta)$, which is *independent* of ε for any fixed $\alpha, \beta > 0$. This perturbation simultaneously enables a *single-loop* primal–dual update and achieves $\mathcal{O}(\varepsilon^{-3})$ stochastic oracle complexity for nonlinear nonconvex constrained optimization, improving over the existing $\mathcal{O}(\varepsilon^{-4})$ barrier for this setting.

1.2 Our Contributions

Our main contributions are summarized as follows.

- We propose a *single-loop* stochastic perturbed proximal ALM for nonlinear equality-constrained nonconvex optimization. The method combines a primal proximal step, a dual smoothing step, and a closed-form multiplier update.
- We introduce a dual perturbation mechanism that yields the effective penalty parameter $\rho = \alpha/(1+\alpha\beta)$, which is constant for fixed (α, β) and independent of the target accuracy ε.
- We establish non-asymptotic convergence guarantees in terms of KKT-type residuals and prove an oracle complexity of $\mathcal{O}(\varepsilon^{-3})$, improving over the $\mathcal{O}(\varepsilon^{-4})$ complexity level of existing single-loop stochastic ALM analyses for nonlinear constraints.
- The framework avoids an increasing-penalty schedule and keeps each iteration simple, which improves practical implementability.

1.3 Notation

Throughout this paper, $\|\cdot\|$ denotes the Euclidean norm and $\langle\cdot,\cdot\rangle$ denotes the Euclidean inner product. We use $\mathbb{E}_\xi[\cdot]$ for expectation with respect to ξ. Let $[p] := \{1,\dots,p\}$, $\mathbb{N} := \{0,1,2,\dots\}$, and $\overline{\mathbb{R}} := \mathbb{R}\cup\{+\infty\}$. For $A \in \mathbb{R}^{m\times n}$, $\|A\|$ and $\sigma_{\min}(A)$ denote the largest and smallest singular values of A, respectively. For $X \subseteq \mathbb{R}^n$, δ_X denotes its indicator function and $\operatorname{dist}(x, X) := \inf_{y\in X}\|x-y\|$. For complexity notation, we write $K = \mathcal{O}(\varepsilon)$ if $K \le B\varepsilon$ for some $B > 0$, and $K = \Theta(\varepsilon)$ if $b\varepsilon \le K \le B\varepsilon$ for some constants $0 < b < B < +\infty$.

1.4 Organization

The rest of the paper is organized as follows. Section 2 introduces the assumptions and the stochastic KKT criterion. Section 3 presents the perturbed proximal ALM formulation and the single-loop algorithm. Section 4 establishes stochastic complexity and residual bounds. Section 5 reports numerical experiments. Additional technical lemmas are provided in Appendix A.

2 Preliminaries and Assumptions

2.1 Main Assumptions and stationarity

In this subsection, we formally state our assumptions for Problem (**P**) and the stationarity for problem.

Assumption 1 (Bounded residual and proximal regularity) *There exists a constant $\Delta > 0$ such that, for any $k \geq 0$,*

$$\|\boldsymbol{c}(\boldsymbol{x}^k)\| \leq \left(\frac{4\Delta}{\rho}\right)^{1/2}, \qquad \|\boldsymbol{\mu}^k\| \leq (\rho\Delta)^{1/2}.$$

Moreover, $g : \mathbb{R}^n \to \overline{\mathbb{R}}$ is proper and lower semicontinuous with

$$\mathcal{X} = \mathrm{dom} g := \{\boldsymbol{x} \in \mathbb{R}^n \mid g(\boldsymbol{x}) < +\infty\},$$

and g is proximal-friendly: for any $\boldsymbol{z} \in \mathbb{R}^n$ and any $t > 0$, the subproblem

$$\min_{\boldsymbol{x} \in \mathbb{R}^n} \; g(\boldsymbol{x}) + \frac{1}{2t}\|\boldsymbol{x} - \boldsymbol{z}\|^2 \tag{2.1}$$

is solvable.

Assumption 2 *Function $f(\cdot)$ is L_f-smooth over $\mathcal{X}$. The mapping $\boldsymbol{c}(\cdot) : \mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable over $\mathcal{X}$. There exist constants $C, G, L_c > 0$ such that for any $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$,*

$$\|\boldsymbol{c}(\boldsymbol{x})\| \leq C, \quad \|\boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{c}(\boldsymbol{y})\| \leq G\|\boldsymbol{x} - \boldsymbol{y}\|, \tag{2.2a}$$

$$\|J_{\boldsymbol{c}}(\boldsymbol{x})\| \leq G, \quad \|J_{\boldsymbol{c}}(\boldsymbol{x}) - J_{\boldsymbol{c}}(\boldsymbol{y})\| \leq L_c\|\boldsymbol{x} - \boldsymbol{y}\|. \tag{2.2b}$$

Assumption 3 *For any $\boldsymbol{x}$, there exists $\sigma > 0$ such that*

$$\begin{aligned} &\mathbb{E}_\xi[\nabla \mathbf{f}(\boldsymbol{x}, \boldsymbol{\xi})] = \nabla f(\boldsymbol{x}), \\ &\|\nabla \mathbf{f}(\boldsymbol{x}, \boldsymbol{\xi}) - \nabla f(\boldsymbol{x})\| \leq \sigma, \quad \forall \xi \end{aligned} \tag{2.3}$$

We also introduce a standard assumption, also known as the mean-squared smoothness assumption, widely used in variance reduction methods.

Assumption 4 *For almost every $\boldsymbol{\xi} \in \Xi$, $f(\cdot, \boldsymbol{\xi})$ is differentiable and for any $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$,*

$$\mathbb{E}_{\boldsymbol{\xi}}\left[\|\nabla f(\boldsymbol{x}; \boldsymbol{\xi}) - \nabla f(\boldsymbol{y}; \boldsymbol{\xi})\|^2\right] \leq L_f^2\|\boldsymbol{x} - \boldsymbol{y}\|^2.$$

2.2 Stochastic KKT Point

Definition 1 (Stochastic KKT point) *A point $\boldsymbol{x}^* \in \mathcal{X}$ is called a* stationary point *of Problem (P) if there exists $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ such that the KKT conditions are satisfied:*

$$\mathbf{0} \in \nabla f(\boldsymbol{x}^*) + \partial g(\boldsymbol{x}^*) + J_{\boldsymbol{c}}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}^* + \mathcal{N}_{\mathcal{X}}(\boldsymbol{x}^*), \qquad \boldsymbol{c}(\boldsymbol{x}^*) = \mathbf{0}. \tag{2.4}$$

3 Methodology and Algorithm

We now define the proximal augmented Lagrangian function

$$\mathcal{L}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \underbrace{f(\boldsymbol{x}) + \langle \boldsymbol{\lambda}, \boldsymbol{c}(\boldsymbol{x})\rangle + \frac{\rho}{2}\|\boldsymbol{c}(\boldsymbol{x})\|^2}_{\varphi_\rho(\boldsymbol{x}, \boldsymbol{\lambda})} + g(\boldsymbol{x}) - \frac{1}{2\rho}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2. \tag{3.1}$$

3.1 Algorithm

Update of $\boldsymbol{x}$. We adopt stochastic proximal descent for the primal variable $\boldsymbol{x}$.

Since the computation of exact gradient information of f is costly and sometimes even prohibitive, we assume that a stochastic gradient at any inquiry point can be accessed. For a mini-batch sample B_k at iteration k, let

$$\nabla \mathbf{f}(\boldsymbol{x}, B_k) := \frac{1}{|B_k|}\sum_{\xi \in B_k} \nabla \mathbf{f}(\boldsymbol{x}, \xi).$$

Then, at the current iterate $\boldsymbol{x}^k$, we compute the stochastic gradient estimator by

$$\boldsymbol{d}^k = \nabla \mathbf{f}(\boldsymbol{x}^k, B_k) + (1-\gamma_{k-1})\big(\boldsymbol{d}^{k-1} - \nabla \mathbf{f}(\boldsymbol{x}^{k-1}, B_k)\big), \qquad k \geq 1, \tag{3.2}$$

with initialization

$$\boldsymbol{d}^0 = \nabla \mathbf{f}(\boldsymbol{x}^0, B_0). \tag{3.3}$$

Here, the same mini-batch B_k is used to evaluate both $\nabla \mathbf{f}(\boldsymbol{x}^k, B_k)$ and $\nabla \mathbf{f}(\boldsymbol{x}^{k-1}, B_k)$.

We define $\tilde{\nabla}\psi_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) := \boldsymbol{d}^k + J_{\boldsymbol{c}}(\boldsymbol{x}^k)^\top \big(\boldsymbol{\lambda}^k + \rho\, \boldsymbol{c}(\boldsymbol{x}^k)\big)$ as the stochastic gradient estimator of $\nabla \varphi_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k)$.

$$\boldsymbol{x}^{k+1} \in \operatorname*{argmin}_{\boldsymbol{x}\in\mathcal{X}} \left\{ g(\boldsymbol{x}) + \langle \tilde{\nabla}\psi_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k),\ \boldsymbol{x} - \boldsymbol{x}^k \rangle + \frac{1}{2\eta_k}\|\boldsymbol{x} - \boldsymbol{x}^k\|^2 \right\}. \tag{3.4}$$

Update of $\boldsymbol{\mu}$. We perform the dual auxiliary variable $\boldsymbol{\mu}$ via gradient ascent step:

$$\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \theta_k\big(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\big), \tag{3.5}$$

with a diminishing step-size

$$\theta_k = \min\left\{\theta_0,\ \frac{\rho\delta_k}{\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 + 1}\right\}, \tag{3.6}$$

and δ_k satisfies

$$\delta_0 \in (0, 1], \qquad \lim_{k\to\infty} \delta_k = 0, \qquad \text{and} \qquad \sum_{k=0}^{\infty} \delta_k = +\infty. \tag{3.7}$$

Update of $\boldsymbol{\lambda}$. We perform the dual variable $\boldsymbol{\lambda}$ update by exactly minimizing

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\mu}^{k+1} + \rho\, \boldsymbol{c}(\boldsymbol{x}^{k+1}). \tag{3.8}$$

Algorithm 1 Linearized DualProximal ALM

1: **Input:** $(\boldsymbol{x}^0)$, $\boldsymbol{\lambda}^0 = \boldsymbol{\mu}^0 = \mathbf{0}$ step-size sequence $\{\eta_k\}_{k\geq 0}$, momentum sequence $\{\gamma_k\}_{k\geq 0}$, and sequence $\{\delta_k\}_{k\geq 0}$
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Update $\boldsymbol{x}^{k+1}$ by (3.4).
4: Update $\boldsymbol{\mu}^{k+1}$ by (3.5).
5: Update $\boldsymbol{\lambda}^{k+1}$ by (3.8).
6: **end for**
7: **Output:** $\boldsymbol{x}^R$

4 Auxiliary Lemmas

We first establish some results on the boundedness of multipliers, the smoothness of φ_ρ, and basic relations on $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$.

Lemma 1 *Let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumption 2, for any $k \geq 0$, it holds that*

$$\|\boldsymbol{\mu}^{k+1}\| \leq \rho C \sum_{j=0}^{k} \theta_j, \tag{4.1}$$

$$\|\boldsymbol{\lambda}^{k+1}\| \leq \rho C \left(1 + \sum_{j=0}^{k} \theta_j\right), \tag{4.2}$$

where C defined in Assumption 2 and rho are independent of K.

Proof 1 *From the $\boldsymbol{\lambda}$-update (3.8),*

$$\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k = \rho\, \boldsymbol{c}(\boldsymbol{x}^k), \qquad \forall k \geq 0.$$

Substituting this identity into the $\boldsymbol{\mu}$-update (3.5) gives

$$\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \rho\theta_k\, \boldsymbol{c}(\boldsymbol{x}^k).$$

Summing from $j = 0$ to k and using $\boldsymbol{\mu}^0 = \mathbf{0}$, we obtain

$$\boldsymbol{\mu}^{k+1} = \rho \sum_{j=0}^{k} \theta_j\, \boldsymbol{c}(\boldsymbol{x}^j).$$

Hence, by $\|\boldsymbol{c}(\boldsymbol{x})\| \leq C$ from Assumption 2,

$$\|\boldsymbol{\mu}^{k+1}\| \leq \rho \sum_{j=0}^{k} \theta_j \|\boldsymbol{c}(\boldsymbol{x}^j)\| \leq \rho C \sum_{j=0}^{k} \theta_j.$$

Plugging this into (3.8),

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\mu}^{k+1} + \rho\, \boldsymbol{c}(\boldsymbol{x}^{k+1}) = \rho \sum_{j=0}^{k} \theta_j\, \boldsymbol{c}(\boldsymbol{x}^j) + \rho\, \boldsymbol{c}(\boldsymbol{x}^{k+1}).$$

Hence, by $\|\boldsymbol{c}(\boldsymbol{x})\| \leq C$ from Assumption 2,

$$\|\boldsymbol{\lambda}^{k+1}\| \leq \rho \sum_{j=0}^{k} \theta_j \|\boldsymbol{c}(\boldsymbol{x}^j)\| + \rho \|\boldsymbol{c}(\boldsymbol{x}^{k+1})\| \leq \rho C \left(1 + \sum_{j=0}^{k} \theta_j\right).$$

This proves (4.1) and (4.2).

Lemma 2 (Smoothness of φ_ρ) *Under Assumptions 2 and Lemma 1, for any $k \geq 0$, the function $\varphi_\rho(\cdot, \boldsymbol{\lambda}^k)$ is L_φ^k-smooth, i.e.,*

$$\|\nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}, \boldsymbol{\lambda}^k) - \nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{y}, \boldsymbol{\lambda}^k)\| \leq L_\varphi^k \|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}, \tag{4.3}$$

where

$$L_\varphi^k := L_f + \rho L_c C \left(2 + \sum_{j=0}^{k-1} \theta_j\right) + \rho G^2, \tag{4.4}$$

with the convention $\sum_{j=0}^{-1} \theta_j = 0$.

Proof 2 *By the definition of φ_ρ,*

$$\varphi_\rho(\boldsymbol{x}, \boldsymbol{\lambda}^k) = f(\boldsymbol{x}) + \langle \boldsymbol{\lambda}^k, \boldsymbol{c}(\boldsymbol{x}) \rangle + \frac{\rho}{2} \|\boldsymbol{c}(\boldsymbol{x})\|^2,$$

and hence

$$\nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}, \boldsymbol{\lambda}^k) = \nabla f(\boldsymbol{x}) + J_{\boldsymbol{c}}(\boldsymbol{x})^\top \boldsymbol{\lambda}^k + \rho J_{\boldsymbol{c}}(\boldsymbol{x})^\top \boldsymbol{c}(\boldsymbol{x}).$$

Therefore,

$$\begin{aligned} &\nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}, \boldsymbol{\lambda}^k) - \nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{y}, \boldsymbol{\lambda}^k) \\ &= \big(\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\big) + \big(J_{\boldsymbol{c}}(\boldsymbol{x}) - J_{\boldsymbol{c}}(\boldsymbol{y})\big)^\top \boldsymbol{\lambda}^k \\ &\quad + \rho \Big(J_{\boldsymbol{c}}(\boldsymbol{x})^\top \boldsymbol{c}(\boldsymbol{x}) - J_{\boldsymbol{c}}(\boldsymbol{y})^\top \boldsymbol{c}(\boldsymbol{y})\Big). \end{aligned}$$

Taking norms and using the triangle inequality, we obtain

$$\begin{aligned}&\|\nabla_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{x},\boldsymbol{\lambda}^k)-\nabla_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{y},\boldsymbol{\lambda}^k)\|\\ \le\ &\|\nabla f(\boldsymbol{x})-\nabla f(\boldsymbol{y})\|+\|(J_{\boldsymbol{c}}(\boldsymbol{x})-J_{\boldsymbol{c}}(\boldsymbol{y}))^\top\boldsymbol{\lambda}^k\|\\ &+\rho\|J_{\boldsymbol{c}}(\boldsymbol{x})^\top\boldsymbol{c}(\boldsymbol{x})-J_{\boldsymbol{c}}(\boldsymbol{y})^\top\boldsymbol{c}(\boldsymbol{y})\|.\end{aligned}$$

We bound the three terms separately.

Since f is L_f-smooth on $\mathcal{X}$,

$$\|\nabla f(\boldsymbol{x})-\nabla f(\boldsymbol{y})\|\le L_f\|\boldsymbol{x}-\boldsymbol{y}\|.$$

By the Lipschitz continuity of $J_{\boldsymbol{c}}$,

$$\|(J_{\boldsymbol{c}}(\boldsymbol{x})-J_{\boldsymbol{c}}(\boldsymbol{y}))^\top\boldsymbol{\lambda}^k\|\le\|J_{\boldsymbol{c}}(\boldsymbol{x})-J_{\boldsymbol{c}}(\boldsymbol{y})\|\,\|\boldsymbol{\lambda}^k\|\le L_c\|\boldsymbol{\lambda}^k\|\,\|\boldsymbol{x}-\boldsymbol{y}\|.$$

Also,

$$\begin{aligned}&J_{\boldsymbol{c}}(\boldsymbol{x})^\top\boldsymbol{c}(\boldsymbol{x})-J_{\boldsymbol{c}}(\boldsymbol{y})^\top\boldsymbol{c}(\boldsymbol{y})\\ =\ &J_{\boldsymbol{c}}(\boldsymbol{x})^\top\big(\boldsymbol{c}(\boldsymbol{x})-\boldsymbol{c}(\boldsymbol{y})\big)+\big(J_{\boldsymbol{c}}(\boldsymbol{x})-J_{\boldsymbol{c}}(\boldsymbol{y})\big)^\top\boldsymbol{c}(\boldsymbol{y}),\end{aligned}$$

hence

$$\begin{aligned}&\rho\|J_{\boldsymbol{c}}(\boldsymbol{x})^\top\boldsymbol{c}(\boldsymbol{x})-J_{\boldsymbol{c}}(\boldsymbol{y})^\top\boldsymbol{c}(\boldsymbol{y})\|\\ \le\ &\rho\|J_{\boldsymbol{c}}(\boldsymbol{x})\|\,\|\boldsymbol{c}(\boldsymbol{x})-\boldsymbol{c}(\boldsymbol{y})\|+\rho\|J_{\boldsymbol{c}}(\boldsymbol{x})-J_{\boldsymbol{c}}(\boldsymbol{y})\|\,\|\boldsymbol{c}(\boldsymbol{y})\|\\ \le\ &\rho G^2\|\boldsymbol{x}-\boldsymbol{y}\|+\rho L_cC\|\boldsymbol{x}-\boldsymbol{y}\|.\end{aligned}$$

Combining the above estimates,

$$\|\nabla_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{x},\boldsymbol{\lambda}^k)-\nabla_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{y},\boldsymbol{\lambda}^k)\|\le\Big(L_f+L_c(\|\boldsymbol{\lambda}^k\|+\rho C)+\rho G^2\Big)\|\boldsymbol{x}-\boldsymbol{y}\|.$$

This proves the first expression in (4.4). From Lemma 1,

$$\|\boldsymbol{\lambda}^k\|\le\rho C\left(1+\sum_{j=0}^{k-1}\theta_j\right),$$

which implies

$$L_\varphi^k\le L_f+\rho L_cC\left(2+\sum_{j=0}^{k-1}\theta_j\right)+\rho G^2.$$

This completes the proof. □

Lemma 3 (Basic relations on $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$) *Let $\{(\boldsymbol{x}^k,\boldsymbol{\lambda}^k,\boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumption 2, the following relations hold for any $k\ge0$:*

$$\|\boldsymbol{\mu}^{k+1}-\boldsymbol{\mu}^k\|^2=\theta_k^2\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2\le\frac{\rho^2\delta_k^2}{4},\tag{4.5}$$

$$\|\boldsymbol{\mu}^{k+1}-\boldsymbol{\lambda}^k\|^2=(1-\theta_k)^2\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2,\tag{4.6}$$

$$\theta_k\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2\le\rho\delta_k,\tag{4.7}$$

$$\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^k\|^2\le2\rho^2L_c^2\,\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2+\frac{\rho^2\delta_k^2}{2}.\tag{4.8}$$

Proof 3 ***Relation*** **(4.5).** *From the* $\boldsymbol{\mu}$*-update* (3.5)*,* $\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k = \theta_k(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k)$*, so*

$$\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|^2 = \theta_k^2\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 \le \frac{\rho^2\delta_k^2}{\left(\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 + 1\right)^2} \cdot \|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 \le \frac{\rho^2\delta_k^2}{4},$$

where the last inequality uses $\frac{t}{(t+1)^2} \le \frac{1}{4}$ *for* $t = \|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 \ge 0$*.*

Relation **(4.7).** *By definition* (3.6)*,* $\theta_k \le \frac{\rho\delta_k}{\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 + 1}$*, hence*

$$\theta_k\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 \le \frac{\rho\delta_k\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2}{\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 + 1} \le \rho\delta_k.$$

Relation **(4.6).** $\boldsymbol{\mu}^{k+1} - \boldsymbol{\lambda}^k = \boldsymbol{\mu}^k - \boldsymbol{\lambda}^k + \theta_k(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k) = (1 - \theta_k)(\boldsymbol{\mu}^k - \boldsymbol{\lambda}^k)$*, so* $\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\lambda}^k\|^2 = (1 - \theta_k)^2\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2$*.*

Relation **(4.8).** *From the* $\boldsymbol{\lambda}$*-update* (3.8)*,* $\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k = (\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k) + \rho\big(\boldsymbol{c}(\boldsymbol{x}^{k+1}) - \boldsymbol{c}(\boldsymbol{x}^k)\big)$*. Using* $(a+b)^2 \le 2(a^2+b^2)$ *and* L_c*-smoothness of* $\boldsymbol{c}$*:*

$$\begin{aligned}\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2 &\le 2\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|^2 + 2\rho^2\|\boldsymbol{c}(\boldsymbol{x}^{k+1}) - \boldsymbol{c}(\boldsymbol{x}^k)\|^2 \\ &\le \frac{\rho^2\delta_k^2}{2} + 2\rho^2 L_c^2\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2.\end{aligned}$$

This proves (4.8)*.* □

For simplicity, we introduce the gradient error estimator

$$\begin{aligned}\boldsymbol{e}^k &:= \nabla_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) - \tilde{\nabla}\psi_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) \\ &= \nabla f(\boldsymbol{x}^k) - \boldsymbol{d}^k. \end{aligned} \tag{4.9}$$

We then prove the approximate sufficient descent property of the augmented Lagrangian function $\mathcal{L}_\rho$ along the iterates generated by Algorithm 1.

Lemma 4 (Sufficient Descent) *Let* $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ *be the sequence generated by Algorithm 1. Under Assumptions 2, the following inequality holds for any* $k \ge 0$*:*

$$\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{\mu}^{k+1}) \;\le\; \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) - C_1^k\,\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 + \frac{\eta_k}{2}\|\boldsymbol{e}^k\|^2 + \hat{\delta}_k, \tag{4.10}$$

where

$$C_1^k := \frac{1}{2\eta_k} - \frac{L_\varphi^k}{2} - \rho L_c^2 > 0, \qquad \hat{\delta}_k := \frac{\rho\delta_k^2}{4} + \delta_k. \tag{4.11}$$

Furthermore, rearranging (4.10) *gives*

$$C_1^k\,\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \le \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) - \mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{\mu}^{k+1}) + \frac{\eta_k}{2}\|\boldsymbol{e}^k\|^2 + \hat{\delta}_k. \tag{4.12}$$

Proof 4 *Decompose the one-step change of* $\mathcal{L}_\rho$ *as*

$$\begin{aligned}&\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{\mu}^{k+1}) - \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) \\ &= \underbrace{\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) - \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)}_{\textit{Part A}} + \underbrace{\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{\mu}^{k+1}) - \mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)}_{\textit{Part B}}.\end{aligned} \tag{4.13}$$

Part A (x-update). *By the* L_φ^k*-smoothness of* $\varphi_\rho(\cdot, \boldsymbol{\lambda}^k)$ *(Lemma 2) and the descent lemma:*

$$\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) - \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) \le \langle\nabla_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k),\, \boldsymbol{x}^{k+1} - \boldsymbol{x}^k\rangle + g(\boldsymbol{x}^{k+1}) - g(\boldsymbol{x}^k) + \frac{L_\varphi^k}{2}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2. \tag{4.14}$$

From the $\boldsymbol{x}$-update optimality condition $-\tilde{\nabla}\psi_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k)-\frac{1}{\eta_k}(\boldsymbol{x}^{k+1}-\boldsymbol{x}^k)\in\partial g(\boldsymbol{x}^{k+1})$, taking $\boldsymbol{x}=\boldsymbol{x}^k$ in the subgradient inequality gives

$$g(\boldsymbol{x}^{k+1})-g(\boldsymbol{x}^k)\le\Big\langle-\tilde{\nabla}\psi_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k)-\tfrac{1}{\eta_k}(\boldsymbol{x}^{k+1}-\boldsymbol{x}^k),\ \boldsymbol{x}^{k+1}-\boldsymbol{x}^k\Big\rangle. \tag{4.15}$$

Substituting (4.15) into (4.14) and using $\tilde{\nabla}\psi_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k)=\nabla_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k)-\boldsymbol{e}^k$:

$$\begin{aligned}&\mathcal{L}_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k,\boldsymbol{\mu}^k)-\mathcal{L}_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k,\boldsymbol{\mu}^k)\\&\le\langle\boldsymbol{e}^k,\,\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\rangle+\Big(\frac{L_\varphi^k}{2}-\frac{1}{\eta_k}\Big)\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2.\end{aligned} \tag{4.16}$$

Applying Young's inequality $\langle\boldsymbol{e}^k,\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\rangle\le\frac{\eta_k}{2}\|\boldsymbol{e}^k\|^2+\frac{1}{2\eta_k}\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2$:

$$\begin{aligned}&\mathcal{L}_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k,\boldsymbol{\mu}^k)-\mathcal{L}_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k,\boldsymbol{\mu}^k)\\&\le-\Big(\frac{1}{2\eta_k}-\frac{L_\varphi^k}{2}\Big)\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2+\frac{\eta_k}{2}\|\boldsymbol{e}^k\|^2.\end{aligned} \tag{4.17}$$

Part B (dual-variable change). *Expanding $\mathcal{L}_\rho$ using definition (3.1):*

$$\textit{Part B}=\langle\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^k,\,\boldsymbol{c}(\boldsymbol{x}^{k+1})\rangle-\frac{1}{2\rho}\Big(\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^{k+1}\|^2-\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2\Big). \tag{4.18}$$

Substituting $\boldsymbol{c}(\boldsymbol{x}^{k+1})=\frac{1}{\rho}(\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^{k+1})$ from (3.8) into the inner product, and applying the polarization identity $\langle a,b\rangle=\frac{1}{2}\|a\|^2+\frac{1}{2}\|b\|^2-\frac{1}{2}\|a-b\|^2$ with $a=\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^k$ and $b=\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^{k+1}$ (so $a-b=\boldsymbol{\mu}^{k+1}-\boldsymbol{\lambda}^k$):

$$\textit{Part B}=\frac{1}{2\rho}\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^k\|^2-\frac{1}{2\rho}\|\boldsymbol{\mu}^{k+1}-\boldsymbol{\lambda}^k\|^2+\frac{1}{2\rho}\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2, \tag{4.19}$$

where $\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^{k+1}\|^2$ cancels exactly. We now bound each term using Lemma 3.

Second and third terms. By (4.6), $\|\boldsymbol{\mu}^{k+1}-\boldsymbol{\lambda}^k\|^2=(1-\theta_k)^2\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2$, so

$$\frac{1}{2\rho}\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2-\frac{1}{2\rho}\|\boldsymbol{\mu}^{k+1}-\boldsymbol{\lambda}^k\|^2=\frac{2\theta_k-\theta_k^2}{2\rho}\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2\le\frac{\theta_k}{\rho}\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2. \tag{4.20}$$

By (4.7), $\theta_k\|\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k\|^2\le\rho\delta_k$, hence this is $\le\delta_k$.

First term. Applying (4.8):

$$\frac{1}{2\rho}\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^k\|^2\le\rho L_c^2\,\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2+\frac{\rho\delta_k^2}{4}. \tag{4.21}$$

Combining the two estimates:

$$\begin{aligned}&\mathcal{L}_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1},\boldsymbol{\mu}^{k+1})-\mathcal{L}_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k,\boldsymbol{\mu}^k)\\&\le\rho L_c^2\,\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2+\frac{\rho\delta_k^2}{4}+\delta_k.\end{aligned} \tag{4.22}$$

Combining Parts A and B. *Adding (4.17) and (4.22):*

$$\begin{aligned}&\mathcal{L}_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1},\boldsymbol{\mu}^{k+1})-\mathcal{L}_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k,\boldsymbol{\mu}^k)\\&\le-\underbrace{\Big(\frac{1}{2\eta_k}-\frac{L_\varphi^k}{2}-\rho L_c^2\Big)}_{=C_1^k}\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2+\frac{\eta_k}{2}\|\boldsymbol{e}^k\|^2+\underbrace{\frac{\rho\delta_k^2}{4}+\delta_k}_{=\hat{\delta}_k}.\end{aligned} \tag{4.23}$$

Condition (4.27) ensures $C_1^k>0$, which completes the proof.

Assume $\gamma_k \equiv \gamma_f \in (0,1)$. It can be derived that, for $k \geq 1$,

$$\boldsymbol{e}^k = \sum_{t=0}^{k} (1-\gamma_f)^t \, \boldsymbol{u}_{k-t},$$

where

$$\boldsymbol{u}_k := \begin{cases} \nabla \mathbf{f}(\boldsymbol{x}^0, B_0) - \nabla f(\boldsymbol{x}^0), & k = 0, \\ \nabla \mathbf{f}(\boldsymbol{x}^k, B_k) - \nabla f(\boldsymbol{x}^k) + (1-\gamma_f)\big(\nabla f(\boldsymbol{x}^{k-1}) - \nabla \mathbf{f}(\boldsymbol{x}^{k-1}, B_k)\big) & k \geq 1. \end{cases} \tag{4.24}$$

Lemma 5 *Under Assumption 2-3. Then for any $K \geq 1$, it holds that with probability at least $1 - 2K\alpha$, that for any $k = 0, \cdots, K-1$, we have*

$$\|\boldsymbol{e}^k\|^2 \;\leq\; 9\log(1/\alpha)\left(\frac{a_f^2}{|B_0|} + \frac{1}{\gamma_f} \max_{1 \leq j \leq k} \Big[\frac{\gamma_f^2 a_f^2}{|B_j|} + \frac{(1-\gamma_f)^2 L_f^2 \|\boldsymbol{x}^j - \boldsymbol{x}^{j-1}\|^2}{|B_j|}\Big]\right),$$

Proof 5 *For $k = 0$, by concentration for mini-batch average,*

$$\|\boldsymbol{u}_0\| \leq 3a_f \sqrt{\frac{\log(1/\alpha)}{|B_0|}} \quad \textit{with probability at least } 1-\alpha.$$

For $k \geq 1$, decompose $\boldsymbol{u}_k$ into a variance part and a drift part:

$$\boldsymbol{u}_k = \underbrace{\frac{1}{|B_k|} \sum_{\xi \in B_k} \Big[\gamma_f\big(\nabla f_\xi(\boldsymbol{x}^k) - \nabla f(\boldsymbol{x}^k)\big) + (1-\gamma_f)\Big(\nabla f_\xi(\boldsymbol{x}^k) - \nabla f_\xi(\boldsymbol{x}^{k-1}) - \big(\nabla f(\boldsymbol{x}^k) - \nabla f(\boldsymbol{x}^{k-1})\big)\Big)\Big]}_{=:\textit{sample fluctuation}} + (1-\gamma_f)\big(\nabla f(\boldsymbol{x}^{k-1}) -$$

Using concentration and smoothness,

$$\|\boldsymbol{u}_k\| \leq 3\gamma_f a_f \sqrt{\frac{\log(1/\alpha)}{|B_k|}} + 6(1-\gamma_f) L_f \|\boldsymbol{x}^k - \boldsymbol{x}^{k-1}\| \sqrt{\frac{\log(1/\alpha)}{|B_k|}}$$

with probability at least $1-\alpha$.

Applying union bound over $k = 0, \ldots, K-1$ gives an event of probability at least $1 - 2K\alpha$ where all above bounds hold simultaneously.

On this event,

$$\boldsymbol{e}^k = \sum_{t=0}^{k} (1-\gamma_f)^t \boldsymbol{u}_{k-t}.$$

By Cauchy-Schwarz (or weighted Jensen),

$$\|\boldsymbol{e}^k\|^2 \leq \Big(\sum_{t=0}^{k} (1-\gamma_f)^t\Big) \sum_{t=0}^{k} (1-\gamma_f)^t \|\boldsymbol{u}_{k-t}\|^2 \leq \frac{1}{\gamma_f} \sum_{t=0}^{k} (1-\gamma_f)^t \|\boldsymbol{u}_{k-t}\|^2.$$

Substituting the high-probability bounds of $\boldsymbol{u}_0, \boldsymbol{u}_1, \ldots, \boldsymbol{u}_k$ yields the claimed inequality.

Lemma 6 (Primal Stationarity Residual Bound) *Let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumptions 2 and 3, define the stationarity residual*

$$\boldsymbol{\zeta}_{\boldsymbol{x}}^{k+1} \in \nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}) + \partial g(\boldsymbol{x}^{k+1}) = \partial_{\boldsymbol{x}} \mathcal{L}_\rho(\boldsymbol{w}^k). \tag{4.25}$$

Then for any $k \geq 0$, Lemma 3 (4.8),

$$\mathbb{E}\big[\|\boldsymbol{\zeta}_{\boldsymbol{x}}^{k+1}\|^2\big] \;\leq\; 4\left[\left(L_\varphi^k + \frac{1}{\eta_k}\right)^2 + 2G^2\rho^2 L_c^2\right] \mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2\big] \;+\; 4\,\mathbb{E}\big[\|\boldsymbol{e}^k\|^2\big] \;+\; 2G^2\rho^2\delta_k^2. \tag{4.26}$$

where $\boldsymbol{e}^k := \nabla f(\boldsymbol{x}^k) - \boldsymbol{d}^k$ is the STORM estimation error.

Lemma 7 *Under Assumptions 2–4, let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be generated by Algorithm 1 with non-increasing step sizes $\{\eta_k\}_{k\geq 0}$ satisfying*

$$(1 - L_{c,\varphi}^k \eta_k)^{-1} \leq 2 \tag{4.27}$$

where $L_{c,\varphi}^k = L_\varphi^k + 2\rho L_c^2$. Then for any $K \geq 1$,

$$\frac{1}{K}\sum_{k=0}^{K-1} \gamma_k \,\mathbb{E}\big[\|\boldsymbol{e}^k\|^2\big] \leq \frac{16\eta_0 L_f^2 \big(\mathcal{L}_\rho^0 - \mathcal{L}_\rho^*\big)}{K} + \frac{2\,\mathbb{E}\big[\|\boldsymbol{e}^1\|^2\big]}{K} + \frac{16\eta_0 L_f^2}{K}\sum_{k=0}^{K-1}\hat{\delta}_k + \frac{1}{K}\sum_{k=0}^{K-1} 4\gamma_k^2\sigma^2, \tag{4.28}$$

where $\mathcal{L}_\rho^* \geq F^*$ is a lower bound on $\mathcal{L}_\rho$.

Lemma 8 (KKT residual stationarity)

$$\begin{aligned}
&\frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}\big[\|\zeta_x^{k+1}\|^2\big] \\
&\leq \left(\frac{\eta_0 L_f^2}{\gamma_K \cdot K} + \frac{1}{\eta_{K-1}\cdot K}\right)\big(\mathcal{L}_\rho^0 - \mathcal{L}^*\big) + \frac{2\,\mathbb{E}\big[\|\mathbf{e}^1\|^2\big]}{\gamma_K \cdot K} + \frac{\displaystyle\sum_{k=0}^{K-1} 4\gamma_k^2\sigma^2}{\gamma_K \cdot K} + \frac{1}{K}\sum_{k=0}^{K-1}\delta_k \\
&\quad + \left[\frac{1}{\eta_{K-1}\cdot K} + \frac{16\eta_0 L_f^2}{\gamma_K \cdot K}\right]\sum_{k=0}^{K-1}\hat{\delta}_k
\end{aligned} \tag{4.29}$$

Lemma 9 (Violation Bound) *Under the same conditions as Lemma 4, with $\delta_k = c/(k+1)$ for some $c > 0$, we have*

$$\frac{1}{T}\sum_{k=0}^{T-1} \mathbb{E}\big[\|\boldsymbol{c}(\boldsymbol{x}^{k+1})\|^2\big] = \mathcal{O}\left(\frac{\log T}{T}\right) \tag{4.30}$$

Proof 6 *Let $r_k := \boldsymbol{\lambda}^k - \boldsymbol{\mu}^k$. By the $\boldsymbol{\lambda}$-update (3.8), we have the identity*

$$\|r_{k+1}\|^2 = \|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2 = \rho^2\|\boldsymbol{c}(\boldsymbol{x}^{k+1})\|^2, \tag{4.31}$$

so it suffices to bound $\frac{1}{T}\sum_{k=0}^{T-1}\mathbb{E}[\|r_{k+1}\|^2]$.

Step 1: Derive the recursion for $\|r_{k+1}\|^2$. *By the triangle inequality and $(a+b)^2 \leq 2(a^2+b^2)$,*

$$\|r_{k+1}\|^2 \leq 2\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2 + 2\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^{k+1}\|^2. \tag{4.32}$$

Applying Lemma 3 (4.6) to the second term and (4.8) to the first:

$$\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^{k+1}\|^2 = (1-\theta_k)^2\|r_k\|^2,$$

$$\mathbb{E}[\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2 \mid \mathcal{F}_k] \leq 2\rho^2 L_c^2\,\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \frac{\rho^2\delta_k^2}{2}.$$

Substituting into (4.32) and using $(1-\theta_k)^2 \leq (1-\theta_k)$ (since $\theta_k \in (0,1]$):

$$\mathbb{E}[\|r_{k+1}\|^2 \mid \mathcal{F}_k] \leq 2(1-\theta_k)\|r_k\|^2 + 4\rho^2 L_c^2\,\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \rho^2\delta_k^2. \tag{4.33}$$

Step 2: Sum the recursion. *Rearranging (4.33):*

$$2\theta_k\|r_k\|^2 \leq 2\|r_k\|^2 - \mathbb{E}[\|r_{k+1}\|^2 \mid \mathcal{F}_k] + 4\rho^2 L_c^2\,\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \rho^2\delta_k^2. \tag{4.34}$$

Taking total expectation and summing over $k = 0, \ldots, T-1$:

$$2\sum_{k=0}^{T-1}\theta_k\,\mathbb{E}[\|r_k\|^2] \leq 2\sum_{k=0}^{T-1}\mathbb{E}[\|r_k\|^2] - \sum_{k=0}^{T-1}\mathbb{E}[\|r_{k+1}\|^2] + 4\rho^2 L_c^2\sum_{k=0}^{T-1}\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \rho^2\sum_{k=0}^{T-1}\delta_k^2. \tag{4.35}$$

Note that the first two sums on the right telescope:

$$2\sum_{k=0}^{T-1}\mathbb{E}[\|r_k\|^2]-\sum_{k=0}^{T-1}\mathbb{E}[\|r_{k+1}\|^2]=\sum_{k=0}^{T-1}\mathbb{E}[\|r_k\|^2]-\sum_{k=1}^{T-1}\mathbb{E}[\|r_k\|^2]+\mathbb{E}[\|r_0\|^2]-\mathbb{E}[\|r_T\|^2]\le\mathbb{E}[\|r_0\|^2]. \quad (4.36)$$

Substituting (4.36) *into* (4.35)*:*

$$2\sum_{k=0}^{T-1}\theta_k\,\mathbb{E}[\|r_k\|^2]\le\|r_0\|^2+4\rho^2L_c^2\sum_{k=0}^{T-1}\mathbb{E}[\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2]+\rho^2\sum_{k=0}^{T-1}\delta_k^2. \quad (4.37)$$

Dividing both sides of (4.37) *by* T*:*

$$\frac{2}{T}\sum_{k=0}^{T-1}\theta_k\,\mathbb{E}[\|r_k\|^2]\le\frac{\|r_0\|^2}{T}+\frac{4\rho^2L_c^2}{T}\sum_{k=0}^{T-1}\mathbb{E}[\|\boldsymbol{x}^{k+1}-\boldsymbol{x}^k\|^2]+\frac{\rho^2}{T}\sum_{k=0}^{T-1}\delta_k^2. \quad (4.38)$$

5 Numerical Experiments

A Supported Lemmas

Lemma 10 (Vector Azuma-Hoeffding Inequality) *Let* $\{\boldsymbol{v}^k\}_{k=1}^K$ *be a vector-valued martingale difference sequence adapted to filtration* $\{\mathcal{F}_k\}_{k=0}^K$*, satisfying*

$$\mathbb{E}[\boldsymbol{v}^k\mid\mathcal{F}_{k-1}]=\mathbf{0},\qquad\|\boldsymbol{v}^k\|\le a_k,\quad k=1,\dots,K. \quad (A.1)$$

Then for any $\delta\in(0,1)$*, with probability at least* $1-\delta$*,*

$$\left\|\sum_{k=1}^K\boldsymbol{v}^k\right\|\le 3\sqrt{\log(1/\delta)\sum_{k=1}^K a_k^2}. \quad (A.2)$$

The above bound is a standard vector-valued Azuma-Hoeffding concentration result.

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